universität freiburg

Machine Learning: Support Vector Machines

Dr. Simon Bultmann, Prof. Dr. Abhinav Valada

Winter Semester 2026-2027
November 9, 2026

robot learning



Outline

1. Linear Classification: Revisited
2. Optimization Problem
3. Non-Linear SVM: The Kernel Trick
4. Conclusion



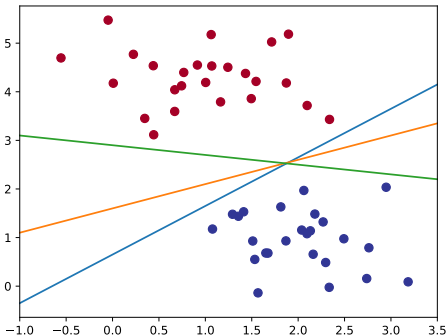
Outline

1. Linear Classification: Revisited
2. Optimization Problem
3. Non-Linear SVM: The Kernel Trick
4. Conclusion



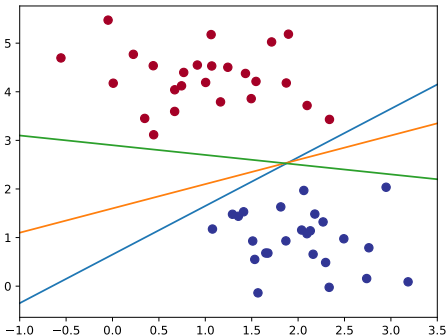
Motivation

We want to classify data using a linear decision boundary:



Motivation

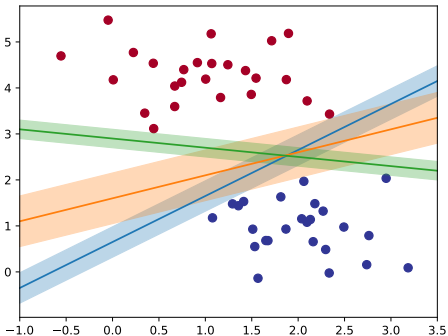
We want to classify data using a linear decision boundary:



Question: Which line best separates the two classes? Why?

Motivation

We want to classify data using a linear decision boundary:

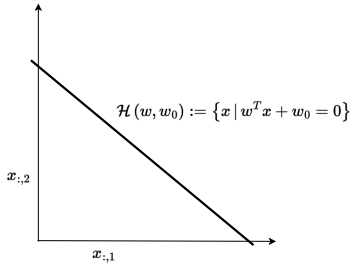


Question: Which line best separates the two classes? Why?

The orange line.

It has the largest **margin** to all the data points and, thus, is more likely to better generalize to unseen data.

Linear Hyperplane



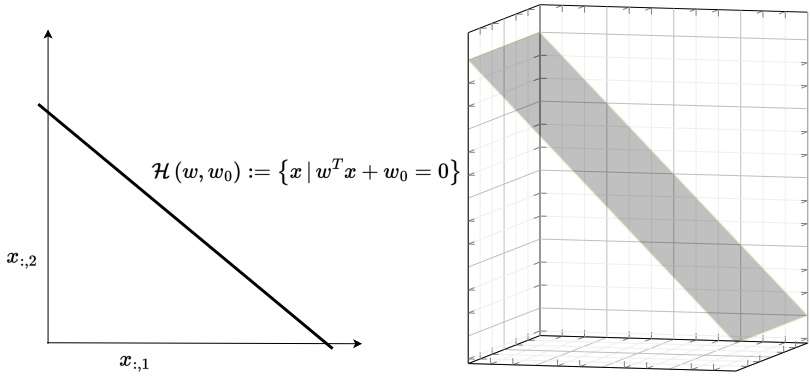
Example linear hyperplanes in 2D.

A linear hyperplane $\mathcal{H}(w, w_0)$ is a sub-space with dimension one less than the dimension of the space $x \in \mathbb{R}^M$.

Question:

What would be the hyperplane of a 3D space?

Linear Hyperplane

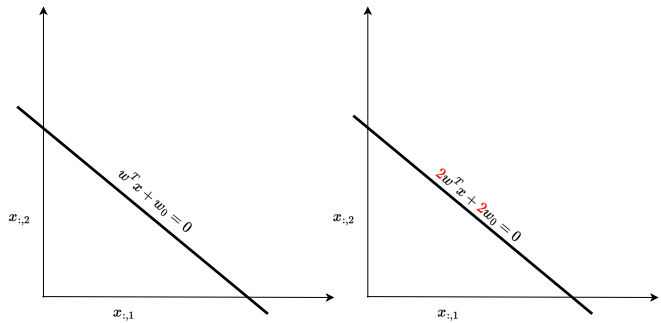


Example linear hyperplanes in 2D and 3D.

Answer: A 2D plane, as shown in the right figure.

Property: the hyperplane divides the space into two parts.

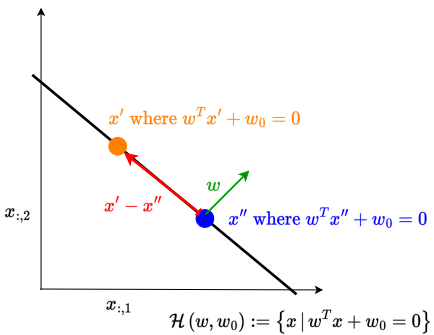
Linear Hyperplane — Scaling w, w_0



Infinitely-many scaled w, w_0 yield the same hyperplane.

$$w^\mathsf{T} x + w_0 = \beta \left(w^\mathsf{T} x + w_0 \right) = (\beta w)^\mathsf{T} x + \beta w_0 = 0, \ \forall \beta \in \mathbb{R}, \beta \neq 0$$

w is orthogonal to the hyperplane



Subtracting the hyperplane equations:

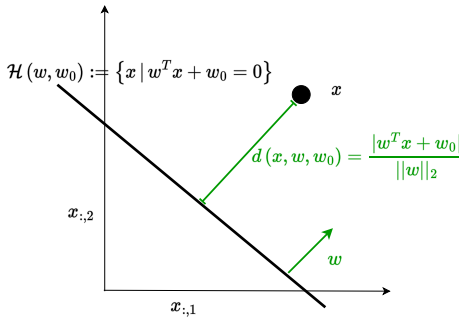
$$w^\top x' + w_0 - (w^\top x'' + w_0) = w^\top (x' - x'') = 0.$$

Using the dot product definition:

$$w^\top (x' - x'') = \|w\|_2 \|x' - x''\|_2 \cos(w,\, x' - x'') = 0.$$

Cosine zero means w is orthogonal to vectors on the plane.

Distance between a hyperplane and a Point

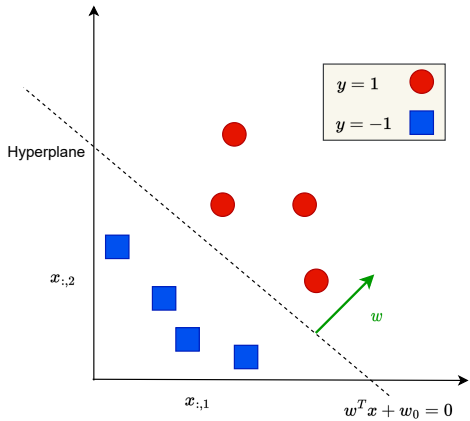


Scaling w, w_0 by any $\beta \in \mathbb{R}, \beta \neq 0$ yields the same distance.

$$\mathrm{d}(x,\, \beta w\,, \beta w_0) = \frac{|\,(\beta w)^\mathsf{T} x + \beta w_0|}{\sqrt{(\beta w)^\mathsf{T}\,(\beta w)^\mathsf{T}}} = \frac{|\beta||w^\mathsf{T} x + w_0|}{\sqrt{\beta^2}\sqrt{w^\mathsf{T} w}} = \mathrm{d}(x,\, w,\, w_0)$$

A linear model for a linearly-separable binary classification

- Features $x \in \mathbb{R}^{N \times M}$, Target $y_i \in \{-1, 1\}^N$



Perceptron: Linear Classification Model

Linear classification problem:

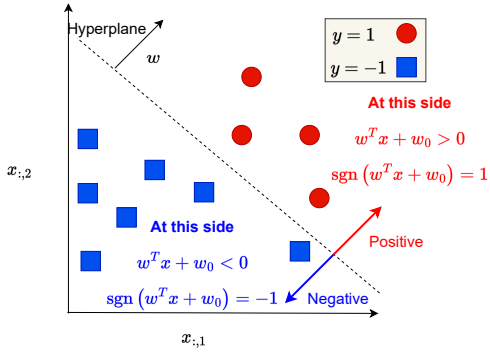
- Model $f(\cdot;\, w, w_0) : \mathbb{R}^M \to \{-1,\, 1\}$ with params $w \in \mathbb{R}^M, w_0 \in \mathbb{R}$

$$\overset{\star}{w} := \arg\min_{w} \sum_{i=1}^{N} \mathcal{L}\left(y_i,\, f(x_i;\, w, w_0)\right)$$

Linear model with a sign function:

$$f(x;\, w) := \operatorname{sign}(w^\intercal x + w_0), \quad \text{with} \quad \operatorname{sign}(x) = \begin{cases} 1 & x > 0 \\ 0 & x = 0 \\ -1 & x < 0 \end{cases}$$

Geometric Interpretation of the Linear Classifier



Classification errors can be formalized as:

$$\forall i: \;\; y_i \left(w^\top x_i + w_0\right) < 0, \text{ or } \forall i: \;\; y_i \; \text{sign}\left(w^\top x_i + w_0\right) = -1$$

Questions to Answer for Yourself / Discuss with Friends

Repetition: Given two vectors $\boldsymbol{w}_a = \alpha \boldsymbol{w}_b$, how are the hyperplanes they represent related?

Repetition: What is the purpose of the $\text{sign}(\cdot)$ function?

Questions to Answer for Yourself / Discuss with Friends

Repetition: Given two vectors $\boldsymbol{w}_a = \alpha \boldsymbol{w}_b$, how are the hyperplanes they represent related?

The two vectors represent the same hyperplane, for any $\alpha \neq 0$

Repetition: What is the purpose of the $\text{sign}(\cdot)$ function?

Questions to Answer for Yourself / Discuss with Friends

Repetition: Given two vectors $\boldsymbol{w}_a = \alpha \boldsymbol{w}_b$, how are the hyperplanes they represent related?

The two vectors represent the same hyperplane, for any $\alpha \neq 0$

Repetition: What is the purpose of the $\text{sign}(\cdot)$ function?

The weights of the model define the decision hyperplane between the classes.

The sign function produces the class labels and restricts the range of the output.

Outline

1 Linear Classification: Revisited

2 Optimization Problem

3 Non-Linear SVM: The Kernel Trick

4 Conclusion



Optimizing the perceptron

Loss over miss-classified instances $y_i \neq \text{sign}\left(w^\mathsf{T} x_i + w_0\right)$ as:

$$\overset{\star}{w} := \arg\min_{w} \sum_{i=1, y_i f(x_i; w)=-1}^{N} -y_i \left(w^\mathsf{T} x_i + w_0\right)$$

Optimizing the perceptron

Loss over miss-classified instances $y_i \neq \text{sign}\left(w^\top x_i + w_0\right)$ as:

$$\overset{\star}{w} := \arg\min_{w} \sum_{i=1, y_i f(x_i;\, w)=-1}^{N} -y_i \left(w^\top x_i + w_0\right)$$

Define the **gradient**: (here $\mathcal{L}_i = \mathcal{L}\left(y_i,\, f(x_i;\, w,\, w_0)\right)$):

$$\frac{\partial \sum_i \mathcal{L}_i}{\partial w} = \sum_{i=1,\ y_i f(x_i;\, w)=-1}^{N} -y_i x_i; \qquad \frac{\partial \sum_i \mathcal{L}_i}{\partial w_0} = \sum_{i=1:\ y_i f(x_i;\, w)=-1}^{N} -y_i$$

Optimizing the perceptron

Loss over miss-classified instances $y_i \neq \text{sign}\left(w^\top x_i + w_0\right)$ as:

$$\overset{\star}{w} := \arg\min_{w} \sum_{i=1, y_i f(x_i;\, w)=-1}^{N} -y_i \left(w^\top x_i + w_0\right)$$

Define the **gradient**: (here $\mathcal{L}_i = \mathcal{L}\left(y_i,\, f(x_i;\, w,\, w_0)\right)$):

$$\frac{\partial \sum_i \mathcal{L}_i}{\partial w} = \sum_{i=1,\ y_i f(x_i;\, w)=-1}^{N} -y_i x_i; \qquad \frac{\partial \sum_i \mathcal{L}_i}{\partial w_0} = \sum_{i=1:\ y_i f(x_i;\, w)=-1}^{N} -y_i$$

Update by step $\eta \in \mathbb{R}_+$ with $\forall (x_i, y_i) : y_i f(x_i; w) = -1$:

$$w^{(t)} \leftarrow w^{(t-1)} + \eta y_i x_i, \qquad w_0^{(t)} \leftarrow w_0^{(t-1)} + \eta y_i$$

Learning algorithm

Algorithm 1: Learning the Perceptron Model

Input: Data $x \in \mathbb{R}^{N \times M}, y_i \in \{-1, 1\}^N$, Learning rate $\eta \in \mathbb{R}_+$

Ensure: $w \in \mathbb{R}^M, w_0 \in \mathbb{R}$

1 $w \sim \mathcal{N}(0, \sigma^2)_M, w_0 \sim \mathcal{N}(0, \sigma^2)$ // Random initial hyperplane

2 errors $\leftarrow 1$

3 **while** errors > 0 **do**

4 errors $\leftarrow 0$

5 **for** $i = 1, \dots, N$ **do**

6 **if** $y_i \neq \text{sign}\left(w^\top x_i + w_0\right)$ **then**

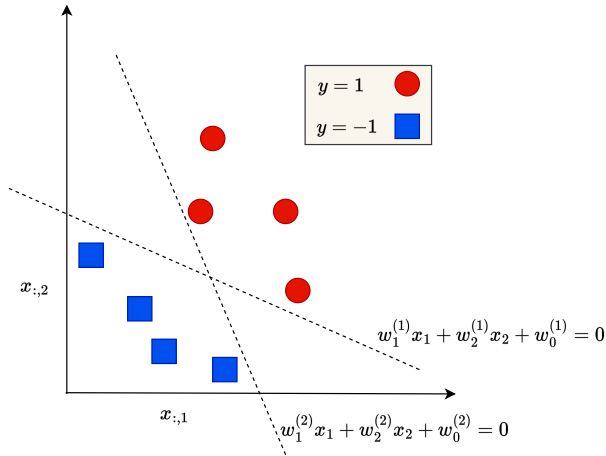
7 errors $\leftarrow$ errors $+ 1$

8 $w \leftarrow w + \eta\, y_i x_i$

9 $w_0 \leftarrow w_0 + \eta\, y_i$

10 **return** w, w_0

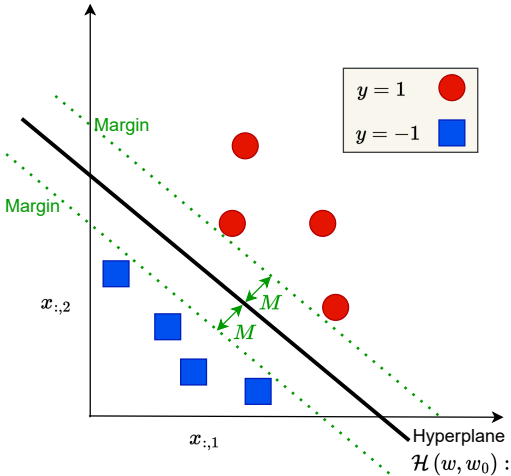
Sub-optimality of the Linear Classifier



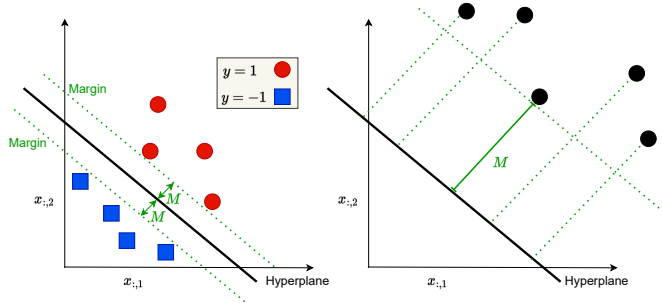
Multiple solution hyperplanes exist. Which one is the optimal?

Interactive SVM demo (KU Leuven) ↗

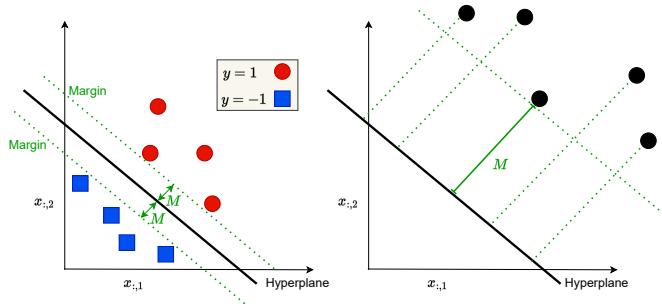
Intuition: Maximum Margin Hyperplane



Margin of a hyperplane to a set of points



Margin of a hyperplane to a set of points



Given points $\{x_1, \ldots, x_N\}$ and plane w, w_0:

$$M(w, w_0) = \min_{x \in \{x_1, \ldots, x_N\}} d(x, w, w_0) = \min_{x \in \{x_1, \ldots, x_N\}} \frac{|w^\top x + w_0|}{\|w\|_2}$$

Maximum margin hyperplane

- Can I optimize the hyperplane directly to yield the maximum margin as follows?

$$\arg\max_{w,\, w_0} \; M(w,\, w_0) = \arg\max_{w,\, w_0} \; \min_{x \in \{x_1, \ldots, x_N\}} \frac{|w^\top x + w_0|}{\|w\|_2}$$

Maximum margin hyperplane

- Can I optimize the hyperplane directly to yield the maximum margin as follows?

$$\arg\max_{w,\, w_0} \; M(w,\, w_0) = \arg\max_{w,\, w_0} \; \min_{x \in \{x_1, \ldots, x_N\}} \frac{|w^\top x + w_0|}{\|w\|_2}$$

  - No, the margin will be increased to infinity.
  - We need to keep the hyperplane between the two classes.

Definition of the Max Margin Hyperplane

Ensure all data points are correctly classified as constraints:

$$\arg\max_{w,\, w_0} \quad \min_{x \in \{x_1, \ldots, x_N\}} \quad \frac{|w^\top x + w_0|}{\|w\|_2}$$

$$\text{s.t. } \forall i: \quad y_i \left(w^\top x_i + w_0\right) \geq 0$$

Get $\|w\|_2$ out of the inner minimization:

$$\arg\max_{w,\, w_0} \quad \frac{1}{\|w\|_2} \quad \min_{x \in \{x_1, \ldots, x_N\}} \quad |w^\top x + w_0|$$

$$\text{s.t. } \forall i: \quad y_i \left(w^\top x_i + w_0\right) \geq 0$$

Simplifying the optimization

Notice there are infinitely many w, w_0 for the same plane:

$$\mathcal{H}\left(w,\, w_0\right) = \left\{x \mid w^\mathsf{T} x + w_0 = 0\right\}$$

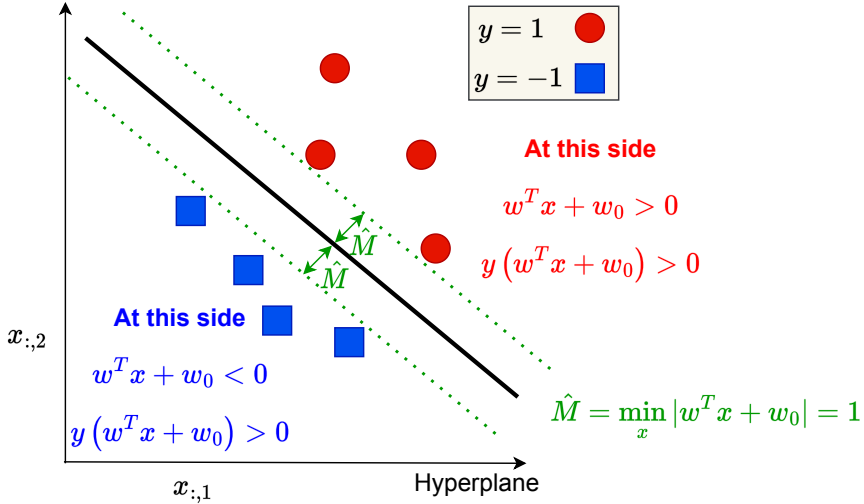
We restrict the infinite space of parameters w, w_0 to a subset:

$$1 = \min_{x \in \{x_1, \ldots, x_N\}} \left| w^\mathsf{T} x + w_0 \right|$$

…in order to simplify our optimization:

$$\begin{aligned} &\arg\max_{w,\, w_0} \quad \frac{1}{\|w\|_2} \\ &\text{s.t.} \quad \forall i: \quad y_i \left(w^\mathsf{T} x_i + w_0\right) \geq 0 \\ &\text{s.t.} \quad \min_{x \in \{x_1, \ldots, x_N\}} \left| w^\mathsf{T} x + w_0 \right| = 1 \end{aligned}$$

Enforcing a margin of 1 unit



Unifying the constraints

The constraints here:

$$\begin{aligned} &\arg\max_{w,\, w_0} \quad \frac{1}{\|w\|_2} \\ &\quad \text{s.t. } \forall i: \quad y_i \left(w^\top x_i + w_0\right) \geq 0 \\ &\quad \text{s.t. } \min_{x \in \{x_1, \ldots, x_N\}} |w^\top x + w_0| = 1 \end{aligned}$$

are equivalent to:

$$\begin{aligned} &\arg\max_{w,\, w_0} \quad \frac{1}{\|w\|_2} \\ &\quad \text{s.t. } \forall i: \quad y_i \left(w^\top x_i + w_0\right) \geq 1 \end{aligned}$$

Converting the objective to a minimization

Convert the maximization of $\|w\|_2$:

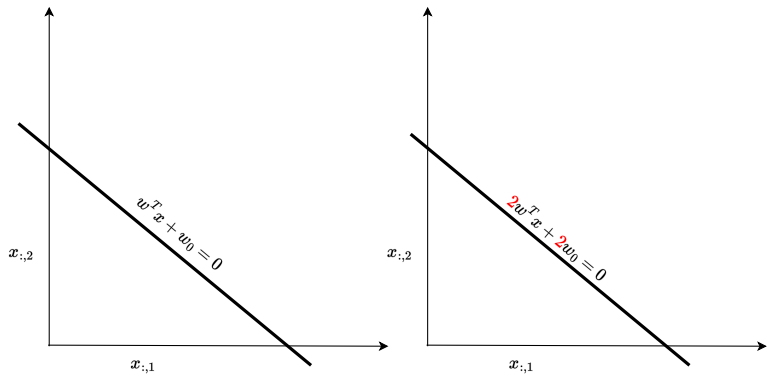
$$\begin{aligned} &\arg\max_{w,\, w_0} \quad \frac{1}{\|w\|_2} \\ &\text{s.t. } \forall i: \quad y_i \left(w^\mathsf{T} x_i + w_0\right) \geq 1 \end{aligned}$$

To a minimization of $w^\mathsf{T} w$:

$$\begin{aligned} &\arg\min_{w,\, w_0} \quad w^\mathsf{T} w \\ &\text{s.t. } \forall i: \quad y_i \left(w^\mathsf{T} x_i + w_0\right) \geq 1 \end{aligned}$$

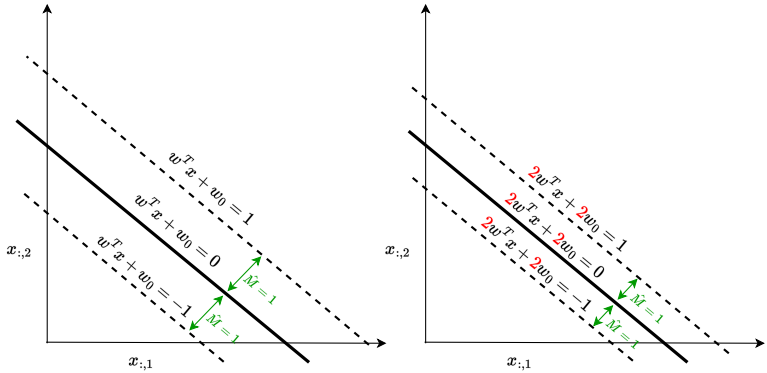
Yielding the objective function for a Linear SVM on a linearly separable task.

Why minimizing $w^\top w$? - Geometric Interpretation



Reminder: by scaling w, w_0, the hyperplane remains the same

Why minimizing $w^\top w$? - Geometric Interpretation



…but the margin decreases inversely proportional to the scaling factor.

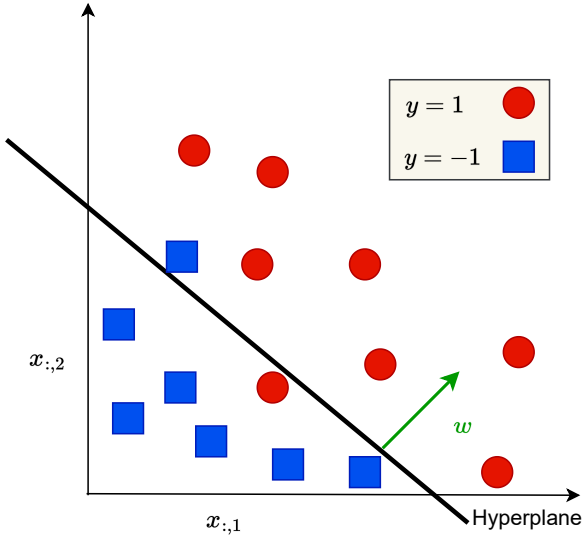
Note: by definition, $\hat{M}$ (also called functional margin) remains 1, but the actual distance M decreases.

Violations to the Linear Separability Assumption

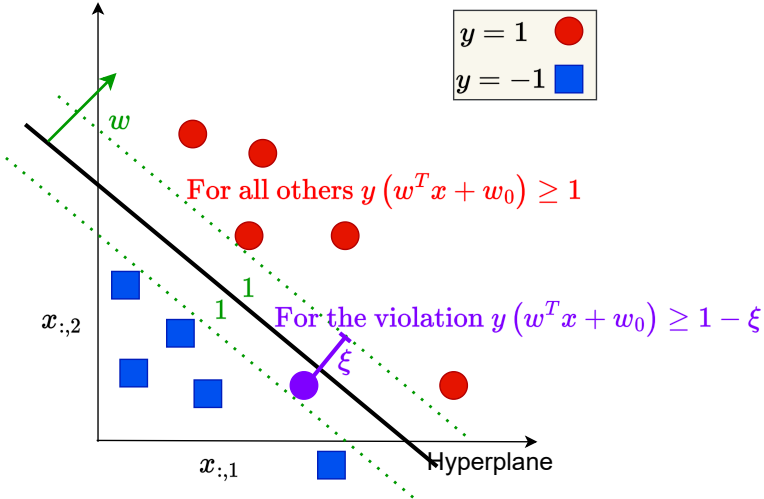
Can we solve w, w_0:

$$\arg\min_{w,\, w_0} \; w^\top w$$
$$\text{s.t. } \forall i: \; y_i \left(w^\top x_i + w_0\right) \geq 1$$

for $\forall i : x_i,\, y_i$ from the dataset on the right?



Slack margin



Tolerate mistakes

…by an amount of violation ξ_i in correctly classifying each y_i, x_i:

$$\begin{aligned} \underset{w,\, w_0}{\arg\min} \;\; & w^\top w \\ \text{s.t. } \forall i: \;\; & y_i \left(w^\top x_i + w_0 \right) \geq 1 - \xi_i \\ \text{s.t. } \forall i: \;\; & \xi_i \geq 0 \end{aligned}$$

Tolerate mistakes

…by an amount of violation ξ_i in correctly classifying each y_i, x_i:

$$\begin{aligned} \underset{w,\, w_0}{\arg\min} \;\; & w^\top w \\ \text{s.t. } \forall i: \;\; & y_i \left(w^\top x_i + w_0 \right) \geq 1 - \xi_i \\ \text{s.t. } \forall i: \;\; & \xi_i \geq 0 \end{aligned}$$

…but the total amount of violations should be minimized:

$$\begin{aligned} \underset{w,\, w_0}{\arg\min} \;\; & w^\top w + C \sum_{i=1}^{N} \xi_i \\ \text{s.t. } \forall i: \;\; & y_i \left(w^\top x_i + w_0 \right) \geq 1 - \xi_i \\ \text{s.t. } \forall i: \;\; & \xi_i \geq 0 \end{aligned}$$

with $C \in \mathbb{R}_+$ controlling the tolerance to violations.

Solving for ξ

From:

$$\begin{aligned} \underset{w,\, w_0}{\arg\min} \;\; & w^\top w + C \sum_{i=1}^{N} \xi_i \\ \text{s.t. } \forall i: \;\; & y_i \left(w^\top x_i + w_0 \right) \geq 1 - \xi_i \\ \text{s.t. } \forall i: \;\; & \xi_i \geq 0 \end{aligned}$$

…we can deduce:

$$\xi_i = \begin{cases} 0 & y_i \left(w^\top x_i + w_0 \right) \geq 1 \\ 1 - y_i \left(w^\top x_i + w_0 \right) & y_i \left(w^\top x_i + w_0 \right) < 1 \end{cases}$$

or rewritten equivalently:

$$\xi_i = \max \left(0,\, 1 - y_i \left(w^\top x_i + w_0 \right) \right)$$

Regularized Hinge Loss Optimization

Replacing ξ we get:

$$\arg\min_{w,\, w_0} \; w^\top w + C \sum_{i=1}^{N} \max\left(0,\, 1 - y_i \left(w^\top x_i + w_0\right)\right)$$

Multiplying by $1/C$ and defining $\lambda = 1/C$:

$$\arg\min_{w, w_0} \; \sum_{i=1}^{N} \max\left(0, 1 - y_i \left(w^\top x_i + w_0\right)\right) + \lambda w^\top w$$

Linear SVM as a Regularized Loss

- Model: $\hat{y}_i(w,\, w_0) = w^\top x_i + w_0, f(x_i;\, w,\, w_0) = \text{sign}\left(\hat{y}_i(w,\, w_0)\right)$
- Loss: $\mathcal{L}\left(y,\, \hat{y}(w,\, w_0)\right) = \max\left(0,\, 1 - y_i\left(w^\top x_i + w_0\right)\right)$
- Regularization $\Omega\left(w\right) = \lambda w^\top w = \lambda \sum_{m=1}^{M} w_m^2$

$$\underset{w,\, w_0}{\arg\min} \sum_{i=1}^{N} \mathcal{L}\left(y_i,\, w^\top x_i + w_0\right) + \lambda \sum_{m=1}^{M} w_m^2$$

Can be solved with Stochastic Gradient Descent exactly like the Logistic Regression.
However, using the sub-gradient of the loss:

$$\frac{d\mathcal{L}(y, \hat{y})}{d\hat{y}} = \frac{d\max(0, 1 - y\hat{y})}{d\hat{y}} = \begin{cases} 0 & y\hat{y} >= 1 \\ -y & y\hat{y} < 1 \end{cases}$$

Dual Optimization

- **Primal form:**
  Constrained optimization of $f(x)$ subject to K constraints $g_1(x) \leq 0, \dots, g_K(x) \leq 0$:

$$\begin{aligned} &\arg\min_x \quad f(x) \\ &\text{s.t. } \forall k: \; g_k(x) \leq 0 \end{aligned}$$

- **Dual Form:**
  An equivalent and simpler form:

$$\begin{aligned} &\arg\max_\alpha \quad \arg\min_x \quad f(x) + \sum_{k=1}^{K} \alpha_k g_k(x) \\ &\text{s.t. } \forall: \; \alpha_k \geq 0 \end{aligned}$$

Primal and Dual SVM formulation

- **Primal SVM form**, notice a constant of $\frac{1}{2}$ is added:

$$\begin{aligned} \arg\min_{w,\, w_0} \quad & \frac{1}{2} w^\top w \\ \text{s.t. } \forall i: \quad & y_i \left(w^\top x_i + w_0 \right) \geq 1 \\ \text{or equivalently,} \forall i: \quad & -y_i \left(w^\top x_i + w_0 \right) + 1 \leq 0 \end{aligned}$$

- **Dual SVM form**:

$$\begin{aligned} \arg\max_{\alpha} \quad \arg\min_{w,\, w_0} \quad & \frac{1}{2} w^\top w - \sum_{i=1}^{N} \alpha_i \left(y_i \left(w^\top x_i + w_0 \right) - 1 \right) \\ \text{s.t. } \forall i: \quad & \alpha_i \geq 0 \end{aligned}$$

Lagrangian Optimization: Intuition

- We want to **minimize an objective** while satisfying **constraints**.
- The Lagrangian combines both using multipliers $\alpha_i \geq 0$:

$$L(w, w_0, \alpha) = \frac{1}{2}\|w\|^2 - \sum_i \alpha_i \left[y_i (w^\top x_i + w_0) - 1 \right].$$

Lagrangian Optimization: Intuition

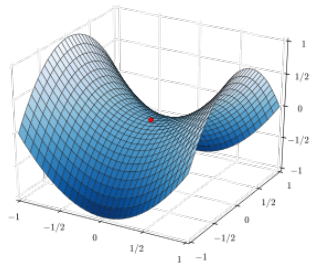
- We want to **minimize an objective** while satisfying **constraints**.
- The Lagrangian combines both using multipliers $\alpha_i \geq 0$:

$$L(w, w_0, \alpha) = \frac{1}{2}\|w\|^2 - \sum_i \alpha_i \left[y_i (w^\top x_i + w_0) - 1 \right].$$

At the optimum, we look for a **saddle point**:

- Minimize w.r.t. w, w_0: optimal model parameters.
- Maximize w.r.t. α: enforce the constraints.

Key intuition: Constraints that determine the optimum have nonzero multipliers (support vectors).



Min–max saddle point

Simplify the Dual form by solving for w

$$\arg\max_{\alpha} \quad \arg\min_{w,\, w_0} \quad \frac{1}{2} w^\top w - \sum_{i=1}^{N} \alpha_i \left(y_i \left(w^\top x_i + w_0 \right) - 1 \right)$$

$$\text{s.t. } \forall i: \quad \alpha_i \geq 0$$

Let $\mathcal{L}(w, w_0, \alpha) = \frac{1}{2} w^\top w - \sum_{i=1}^{N} \alpha_i \left(\left(w^\top x_i + w_0 \right) y_i - 1 \right)$. Then, solve for w, w_0 in terms of α (recall that derivatives at min $= 0$):

$$0 = \frac{\partial \mathcal{L}(w,\, w_0,\, \alpha)}{\partial w} = w - \sum_{i=1}^{N} \alpha_i x_i y_i \qquad w = \sum_{i=1}^{N} \alpha_i x_i y_i$$

$$0 = \frac{\partial \mathcal{L}(w,\, w_0,\, \alpha)}{\partial w_0} = -\sum_{i=1}^{N} \alpha_i y_i \qquad 0 = \sum_{i=1}^{N} \alpha_i y_i$$

Dual SVM Objective

Plugging $w = \sum_{i=1}^{N} \alpha_i x_i y_i$ and setting $0 = \sum_{i=1}^{N} \alpha_i y_i$ to:

$$\arg\max_{\alpha} \quad \arg\min_{w,\, w_0} \quad \frac{1}{2} w^\top w - \sum_{i=1}^{N} \alpha_i \left(y_i \left(w^\top x_i + w_0 \right) - 1 \right)$$
$$\text{s.t. } \forall i: \quad \alpha_i \geq 0$$

yields:

$$\arg\max_{\alpha} \quad \sum_{i=1}^{N} \alpha_i - \frac{1}{2} \sum_{i=1}^{N} \sum_{j=1}^{N} \alpha_i \alpha_j y_i y_j x_i^\top x_j$$
$$\text{s.t. } \sum_{i=1}^{N} \alpha_i y_i = 0, \quad \forall i: \ \alpha_i \geq 0$$

Three Views of the Hard-Margin SVM

A. Primal: optimize the classifier parameters directly

$$\min_{w,w_0} \ \tfrac{1}{2}\|w\|^2 \quad \text{s.t.} \quad y_i(w^\top x_i + w_0) \geq 1 \quad (\forall i).$$

B. Lagrangian: introduce multipliers for the margin constraints

$$L(w, w_0, \alpha) = \tfrac{1}{2}\|w\|^2 - \sum_i \alpha_i \big[y_i(w^\top x_i + w_0) - 1\big], \qquad \alpha_i \geq 0.$$

C. Dual: eliminate w, w_0 using $\nabla_w L = 0$, $\partial L / \partial w_0 = 0$

$$\max_{\alpha} \ \sum_i \alpha_i - \tfrac{1}{2} \sum_{i,j} \alpha_i \alpha_j y_i y_j \, x_i^\top x_j, \quad \text{s.t.} \quad \alpha_i \geq 0, \ \sum_i \alpha_i y_i = 0.$$

Key insight: Only pairwise training-data dot products remain.

Which SVM Formulation Should We Use?

	A. Primal	B. Lagrangian	C. Dual
Variables	w, w_0	w, w_0, α	α
Useful for	Gradient-based training Linear SVMs	Geometric reasoning Active constraints	Kernel methods Similarity-based training
Small problems	Possible	**Often easiest** Use margin equations	Usually more algebra than needed
Computation	Efficient for large linear datasets	Derivation tool, not usually a solver	Gram matrix needs $O(n^2)$ storage

Takeaway: A and C describe the same optimum for separable hard-margin SVMs (strong duality), but they expose different structure. B connects them.

Dual Prediction Model

Remember the linear prediction model:

$$f(x,\, w,\, w_0) = \text{sign}\left(w^\top x + w_0\right)$$

Plugging in $w = \sum_{i=1}^{N} \alpha_i x_i y_i$ leads to:

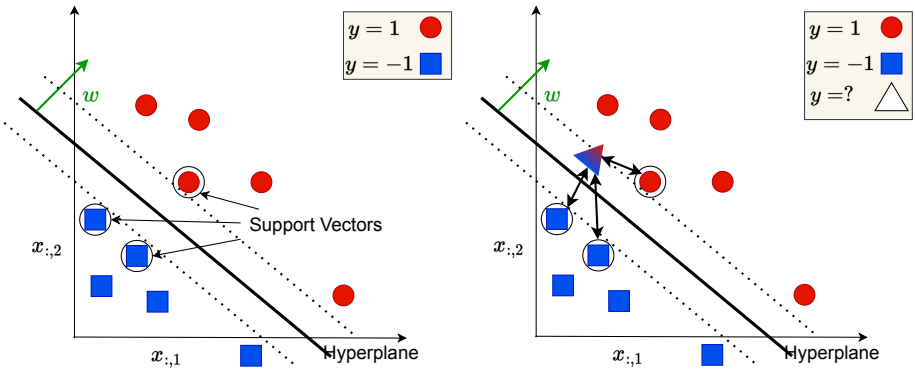
$$f(x,\, \alpha,\, w_0) = \text{sign}\left(\sum_{i=1}^{N} \alpha_i y_i x_i^\top x + w_0\right)$$

Where w_0 is computed as:

$$\forall i : y_i \left(w^\top x_i + w_0\right) = 1 \text{ leads to } \quad \forall i : w_0 = y_i - w^\top x_i$$

Only the instances with $\alpha_i > 0$ matter in the prediction. They are the ”support” vectors/points.

Dual Prediction Model — Inference



In the dual formulation, to predict the label of a new instance, we compute its similarity (dot product) with the support vectors.

Questions to Answer for Yourself / Discuss with Friends

Ponder: What do you think is the effect of the regularization term given by $\Omega(w) = \lambda w^\top w = \lambda \sum_{m=1}^{M} w_m^2$?

Repetition: What are the advantages of the dual SVM formulation?

Questions to Answer for Yourself / Discuss with Friends

Ponder: What do you think is the effect of the regularization term given by $\Omega(w) = \lambda w^\top w = \lambda \sum_{m=1}^{M} w_m^2$?

As seen in the previous lecture, regularization leads to smaller weights and better generalization.

In SVMs, the largest margin is defined by the shortest normal vector w (which still leads to correct classification).

Repetition: What are the advantages of the dual SVM formulation?

Questions to Answer for Yourself / Discuss with Friends

Ponder: What do you think is the effect of the regularization term given by $\Omega(w) = \lambda w^\top w = \lambda \sum_{m=1}^{M} w_m^2$?

As seen in the previous lecture, regularization leads to smaller weights and better generalization.

In SVMs, the largest margin is defined by the shortest normal vector w (which still leads to correct classification).

Repetition: What are the advantages of the dual SVM formulation?

- **Sparse representation:** Only support vectors ($\alpha_i > 0$) contribute to the classifier.
- **Fixed similarities:** Training depends only on dot products between training points ($x_i^\top x_j$), which can be precomputed.
- **Kernel trick:** Dot products can be replaced by kernel functions, enabling nonlinear classification without explicitly computing high-dimensional features.

Outline

1 Linear Classification: Revisited

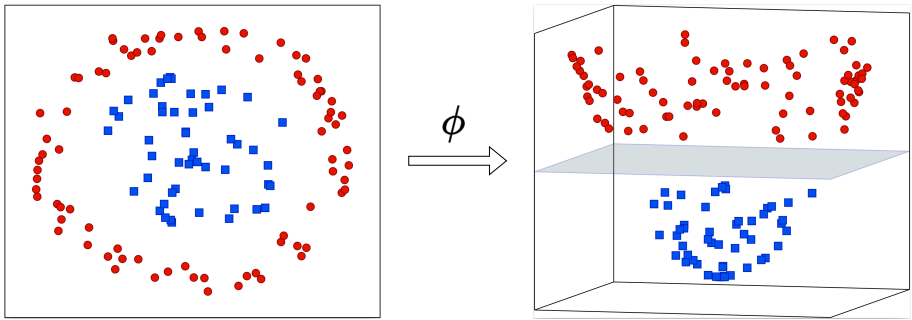
2 Optimization Problem

3 Non-Linear SVM: The Kernel Trick

4 Conclusion



Nonlinear Mapping



By applying a nonlinear mapping $\phi(x)$ to the data, we can make the data linearly separable in a higher dimensional space.

Nonlinear mapping in the optimization

Dual objective:

$$\arg\max_{\alpha} \sum_{i=1}^{N} \alpha_i - \frac{1}{2} \sum_{i=1}^{N} \sum_{j=1}^{N} \alpha_i \alpha_j y_i y_j (\phi(x_i)^\top \phi(x_j))$$

Problem: the dot product $\phi(x_i)^\top \phi(x_j)$ is expensive to compute for high dimensional features.

Solution: kernel functions. A kernel function $K(x_i, x_j)$ is a function that computes the dot product in a higher dimensional space, without explicitly computing the mapping ϕ:

$$K(x_i, x_j) = \phi(x_i)^\top \phi(x_j)$$

Replacing the dot product with a kernel function

Dual objective:

$$\arg\max_{\alpha} \sum_{i=1}^{N} \alpha_i - \frac{1}{2} \sum_{i=1}^{N} \sum_{j=1}^{N} \alpha_i \alpha_j y_i y_j K(x_i, x_j)$$

$$\text{s.t.} \sum_{i=1}^{N} \alpha_i y_i = 0, \quad \forall i : \ \alpha_i \geq 0$$

Dual prediction model:

$$f(x, w, w_0) = \text{sign}\left(\sum_{i=1}^{N} \alpha_i y_i K(x_i, x) + w_0 \right)$$

A nonlinear kernel creates a nonlinear classifier.

Nonlinear kernels yield nonlinear models

$$f(x,\, w,\, w_0) = \operatorname{sign}\left(\sum_{i=1}^{N} \alpha_i y_i K(x_i,\, x) + w_0\right)$$

RBF $K(p,q) = e^{-\gamma(p-q)^2}$, polynomial $K(p,q) = (p^\top q + c)^d$, etc.

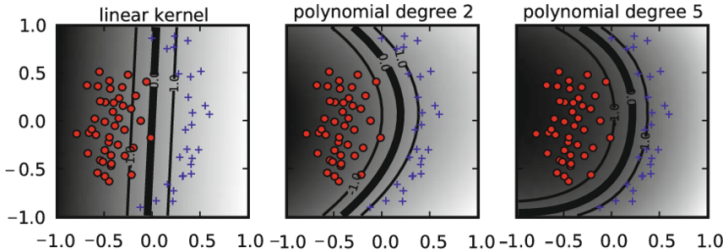


Image source: Asa Ben-Hur et al., 2010

Optimizing the Dual Form

How to find α?

The optimization of the Dual SVM objective (with kernels and slack margins) is not covered in this course.

However, there exist many algorithms for solving the dual formulation. The classic approach is:

- Platt, Sequential Minimal Optimization: A Fast Algorithm for Training Support Vector Machines. [Link]

Questions to Answer for Yourself / Discuss with Friends

Repetition: Why do we use the kernel trick when using non-linear mappings?

Repetition: Can you mention some types of kernels?

Questions to Answer for Yourself / Discuss with Friends

Repetition: Why do we use the kernel trick when using non-linear mappings?

Computing the dot product $\phi(x_i)^\top \phi(x_j)$ is expensive in high dimensional spaces.

A **kernel** function computes the dot product in the higher dimensional space, without computing the mapping $\phi(\cdot)$

Repetition: Can you mention some types of kernels?

Questions to Answer for Yourself / Discuss with Friends

Repetition: Why do we use the kernel trick when using non-linear mappings?

Computing the dot product $\phi(x_i)^\top \phi(x_j)$ is expensive in high dimensional spaces.

A **kernel** function computes the dot product in the higher dimensional space, without computing the mapping $\phi(\cdot)$

Repetition: Can you mention some types of kernels?

- Linear
- Polynomial
- Radial Basis Function (RBF)

Outline

1 Linear Classification: Revisited

2 Optimization Problem

3 Non-Linear SVM: The Kernel Trick

4 Conclusion



Summary

In this lecture we have learned:

- Hyper-planes and the maximum margin principle
- Support Vector Machines (SVM), another learning-based method for classification
- The optimization problem and its dual form
- How to classify data that is not linearly separable using the kernel trick.

Method Recap

Method	Tasks		Complexity		Pros	Cons
	Class.	Regr.	Train / Fit	Inference		
K-Nearest Neighbors	B, MC	✓	—	$\mathrm{O}([k]\, n\, f)$	⊕ Simple and quick ⊕ Can handle noisy data	⊖ Expensive as attributes increase ⊖ Unweighted attributes
Naïve Bayes	B, MC	—	$\mathrm{O}(k\, n\, f)$	$\mathrm{O}(k\, f)$	⊕ Simple ⊕ Useful for large datasets ⊕ Requires little data ⊕ Probabilistic Predictions ⊕ Continuous and discrete data	⊖ Classes must be mutually exclusive ⊖ Assumes normally distr. attributes ⊖ Expects uncorrelated attributes
Linear Regression	—	✓	$\mathrm{O}\left(f^2(n+f)\right)$	$\mathrm{O}(f)$	⊕ Simple and interpretable ⊕ Analytic solution	⊖ Strong, unrealistic assumptions for residuals ⊖ Affected by outliers ⊖ Not scalable, unless done iteratively ⊖ Only for linear data, unless using basis functions
Linear Classification	B	—	$\mathrm{O}\left(f^2(n+f)\right)$	$\mathrm{O}(f)$	⊕ Simple ⊕ Useful for large datasets ⊕ Requires little data	⊖ Only for linearly separable data ⊖ Binary ⊖ Predictions are not probabilistic ⊖ Sensitive to outliers
Logistic Regression	B	—	$\mathrm{O}(n\, k\, f\, i)$	$\mathrm{O}(k\, f)$	⊕ Easy to implement ⊕ No assumptions about indep. variables ⊕ Probabilistic predictions	⊖ Only for linearly separable data ⊖ Binary, unless multinomial ⊖ Inaccurate for highly correlated inputs ⊖ Prone to overfitting

Method Recap (cont)

Method	Tasks		Complexity		Pros	Cons
	Class.	Regr.	Train / Fit	Inference		
SVM	B	—	$\mathrm{O}\left(fn^2\right)$	$\mathrm{O}\left(f\right)$	⊕ Works well when there is a clear margin ⊕ Effective in high dimensions ⊕ Good for $f > n$ ⊕ Memory efficient	⊖ Can't handle overlapping classes ⊖ Bad for large datasets ⊖ Not great for noisy data ⊖ Choosing a good kernel is hard